Unit-4 Dividend policy

Gordon walter

Walter=

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke}$$

D= Dividend
R= Internal rate of return
E= Earning per share
Ke= Equity capital.
P = Market price of share.

Q A company earns ₹ 5/shares is capitalized at a rate of 10%. and has a rate of return on investment @ 18%. According to walter approach what would be the price per share at 25% dividend pay out ratio.

Sol: $P = \frac{D + \frac{r(E-D)}{Ke}}{Ke}$

Here D = 25% of ₹ 5 = ₹1.25
E = 5
r = 0.18
Ke = 0.10

$$= \frac{₹1.25 + \frac{0.18(5-1.25)}{0.10}}{0.10} = \frac{₹1.25 + \frac{0.18 \times 3.75}{0.10}}{0.10} = \frac{₹1.25 + 6.75}{0.1}$$

$$= \frac{8}{0.1} = ₹80$$

Q) The following information is available capitalization rate is 10%. Earnings per share ₹ 50. Assume rate return on investments (i) 12% (ii) 8% (iii) 20%. Show the effect of dividend policy on market price of share apply walter formula. When dividend payout ratio is a) 0% b) 20% c) 40% d) 80% e) 100%.

S) a) Ke = ~~0.1~~ 0.1
E = ₹50
r = 12%
D = 0

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{0 + \frac{0.12(50-0)}{0.1}}{0.1} = 600$$

b) D = 50 × 20% = 10

$$\frac{10 + \frac{0.12(50-10)}{0.1}}{0.1} = ₹580$$

10.12 40. 400

c) D = 50 × 40% = 20

$$= \frac{20 + \frac{0.12(50-20)}{0.1}}{0.1} = ₹560$$

40 48

d) $D = 50 \times 80\% = 40$

$$= \frac{40 + \frac{0.12(50-40)}{0.1}}{0.1} = ₹520$$

e) $D = 50 \times 100\% = 50$

$$= \frac{50 + \frac{0.12(50-50)}{0.1}}{0.1} = ₹500$$

(iii) $r = 8\%$

a) $$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{P = 0 + \frac{0.08(50-0)}{0.1}}{0.1} = ₹400$$

$r = 8\% \Rightarrow 0.08$, $Ke = 0.1$, $D = 50$ of 0%, $E = 50$

b) $r = 8\% \Rightarrow 0.08$, $Ke = 0.1$, $D = 50$ of $20\% = 10$, $E = 50$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{P = 10 + \frac{0.08(50-10)}{0.1}}{0.1} = ₹420$$

c) $r = 0.08\%$, $Ke = 0.1$, $D = 40\ (50 \times 80\%)$, $E = 50$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{40 + \frac{0.08(50-40)}{0.1}}{0.1} = ₹480$$

c) $R = 0.08$, $E = 50$, $D = 20$, $Ke = 0.1$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{20 + \frac{0.08(50-20)}{0.1}}{0.1}$$

e) $R = 0.08$, $E = 50$, $Ke = 0.1$, $D = 50\ (50 \times 10 = 0.1)$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{50 + \frac{0.08(50-50)}{0.1}}{0.1} = ₹500$$

iii) a) $$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke}$$ $r = 0.1$, $E = 50$, $D = 0\ (50 \times 0\%)$, $Ke = 0.1$

$$= \frac{0 + \frac{0.1(50-0)}{0.1}}{0.1} = ₹500$$

b) $r = 0.1$, $E = 50$, $D = 10\ (20\% \times 50)$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{10 + \frac{0.1(50-10)}{0.1}}{0.1} = ₹500$$

c) $R = 0.1$, $E = 50$, $Ke = 0.1$, $D = 50 \times 40\% = 20$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{20 + \frac{0.1(50-20)}{0.1}}{0.1} = ₹500$$

d) $r = 0.1$, $E = 50$, $Ke = 0.1$, $D = 50 \times 80\% = ₹40$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{40 + \frac{0.1(50-40)}{0.1}}{0.1} = ₹500$$

e) $r = 0.1$, $E = 50$, $Ke = 0.1$, $D = 50 \times 100\% = 50$

$$P = \frac{D + \frac{r(E-D)}{Ke}}{Ke} = \frac{50 + \frac{0.1(50-50)}{0.1}}{0.1} = ₹500$$

Gordon Approach

$$P = \frac{E(1-b)}{Ke - br}$$

E = Earnings per share

b = Retention ratio

P = price of shares

Ke = Cost of Equity Capital

br = growth rate

R = rate of return on investment.

Q) The following information is available $r = 10\%$, $E = ₹40$.

	D/P ratio (1-b)	Retention ratio (b)	Cost of equity Ke%
a	20	80	20
b	40	60	18
c	60	40	16
d	80	20	14

(Sol)- D/P ratio = 20% (1-b) → 1-2.8 = 0.2

Retention ratio (b) 80% → 0.8

Cost of Equity (Ke) 20% → 0.2

$$P = \frac{E(1-b)}{Ke-br} = \frac{₹40(1-0.8)}{0.2-0.8\times0.1} = \frac{8}{0.2-0.08} = \frac{8}{0.12} = ₹66.67$$

b) D/P ratio = 40% (1-b)

Retention ratio (b) = 60%

Cost of Equity (Ke) = 80% →

$$P = \frac{E(1-b)}{ke-br} = \frac{₹40(1-0.6)}{0.18-0.6\times0.1} = \frac{16}{0.18-(0.06)} = \frac{16}{0.12} = ₹133.33$$

c) D/P ratio = 60% (1-b)

Retention ratio (b) = 40%

Cost of equity (Ke) = 16%

$$P = \frac{E(1-b)}{Ke-br} = \frac{₹40(1-0.4)}{0.16-(0.4\times0.1)} = \frac{24}{0.16-0.04} = \frac{24}{0.12} = ₹200$$

d) D/P ratio = 80%

retention ratio (b) = 20%

cost of equity (Ke) = 14%

$$P = \frac{E(1-b)}{ke-br} = \frac{40(1-0.2)}{0.14-(0.2\times0.1)} = \frac{32}{0.14-0.02} = \frac{32}{0.12} = ₹266.67$$

Q. Rate of return on investment = 15%, 12% & 10%, Ke = 12%

E = ₹10, b = (a) 0.1% (b) 20% (c) 60%

Sol (i) (a) $P = \frac{E(1-b)}{Ke-br} = \frac{₹10(1-0)}{0.12-(0\times0.15)} = \frac{10}{0.12} = 83.343.$

(b) $P = \frac{E(1-b)}{Ke-br} = \frac{₹10(1-0.2)}{0.12-(0.2\times0.15)} = \frac{8}{0.12-0.03} = \frac{8}{0.09} = ₹88.89$

(c) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0.6)}{0.12-(0.6\times0.15)} = \frac{4}{0.42-0.09} = \frac{4}{0.03} = ₹133.33.$

(ii) (a) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0)}{0.12-(0\times0.12)} = \frac{10}{0.12} = 83.34$

(b) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0.2)}{0.12-(0.2\times0.12)} = \frac{8}{0.12-0.024} = \frac{8}{0.096} = 83.34$

(c) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0.6)}{0.12-(0.6\times0.12)} = \frac{4}{0.12-0.072} = \frac{4}{0.048} = ₹83.34$

(iii) (a) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0)}{0.12(0\times0.1)} = \frac{10}{0.12} = ₹83.34$

b) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0.2)}{0.12-(0.2\times0.1)} = \frac{8}{0.12-0.02} = \frac{8}{0.1} = ₹80$

c) $P = \frac{E(1-b)}{Ke-br} = \frac{10(1-0.6)}{0.12(0.6\times0.10)} = \frac{4}{0.12-0.06} = ₹66.67$